

Data Integration & Correlation

Quick Recap & DIT modules

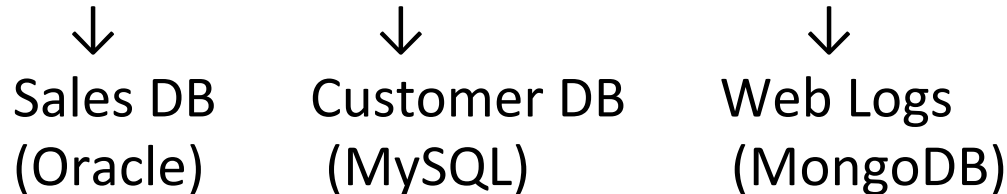
- **What we've covered:**
 - DM introduction & KDD process
 - Descriptive statistics & visualization
 - Data cleaning & outlier detection
- **DIT Modules**
 - **Data Integration Challenges**
 - **Transformation Fundamentals**
 - **Normalization Techniques**
 - **Discretization Methods**

Why is this important? Real-world data rarely comes from a single, clean source!

The Real-World Data Scenario

- **Typical Data Mining Reality**

Multiple Databases → Data Warehouse → Mining



- **Integration Challenges:**

- Different formats (relational, NoSQL, flat files)
- Varying structures and schemas
- Semantic differences
- Temporal inconsistencies
- **Redundant information**

What is Data Integration?

Definition: The process of combining data from different sources into a coherent data store (typically a data warehouse).

- **Key Components:**

- **Schema Integration:** Merging metadata/models
- **Entity Identification:** Matching real-world entities
- **Conflict Resolution:** Handling inconsistencies
- **Redundancy Detection:** Eliminating duplicate information

- **Mathematically:**

Given sources $S_1, S_2, \dots, S_n$ with schemas $\Sigma_1, \Sigma_2, \dots, \Sigma_n$

Goal: Create unified schema Σ and instance I where:

- I contains all information from $S_1 \dots S_n$
- Minimal redundancy
- Consistent representation

Schema Matching Challenge

Problem: Different names for same attributes across databases

- **Example - University Databases:**

Admissions DB: Registrar DB: Housing DB:

- Student_ID	- SID	- ID_Number
- First_Name	- FName	- Given_Name
- Last_Name	- LName	- Surname
- GPA	- Grade_Avg	- Academic_Index

- **Approaches to Resolution:**

- **Manual Specification** (domain expert)
- **Correlation Analysis** (statistical matching)
- **Machine Learning** (learn matching patterns)
- **Ontology-based** (semantic web approaches)

Schema Matching Challenge contd.

Problem: Same entity has different representations across sources

- **Examples:**

- **Customer Records:**

- Source 1: "John Smith, 123 Main St"
 - Source 2: "J. Smith, 123 Main Street"
 - Source 3: "Smith, John, 123 Main St. Apt 4B"

- **Product Matching:**

- "iPhone 13 Pro 256GB Blue"
 - "Apple iPhone 13 Pro 256GB - Sierra Blue"
 - "IPHONE13PRO/BLEUE/256"

- **Solution Techniques:**

- **Record Linkage:** Matching algorithms
 - **Fuzzy Matching:** Handling variations
 - **Unique ID Assignment:** Creating common key

Redundancy & Correlation Analysis

Problem: Same information stored multiple times

- **Detection Methods:**

- **Correlation Coefficient** (for numerical attributes):

$$r = \frac{cov(x, y)}{s_x \cdot s_y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

- $|r| \approx 1$: Strong correlation (potential redundancy)
 - $r \approx 0$: Independent attributes

- **Chi-square Test** (for categorical attributes):

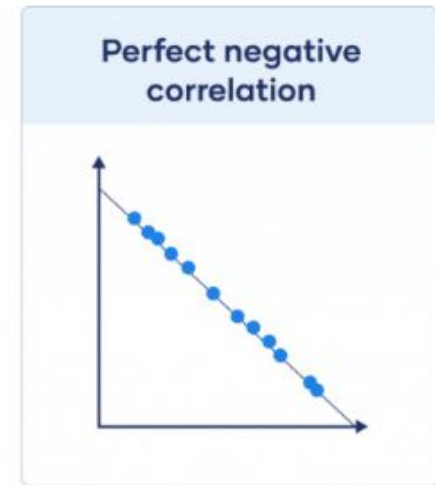
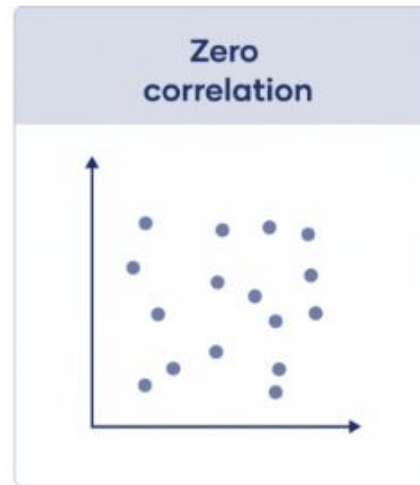
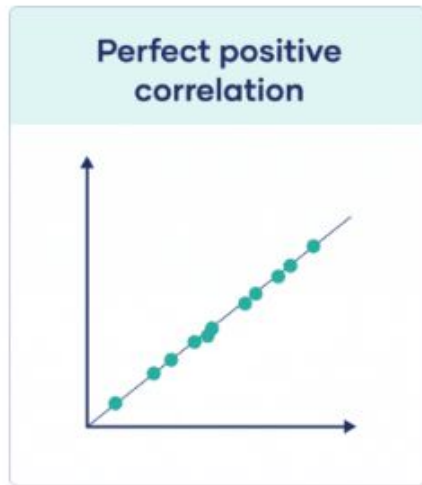
$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

- Tests independence between attributes

Pearson Coeff

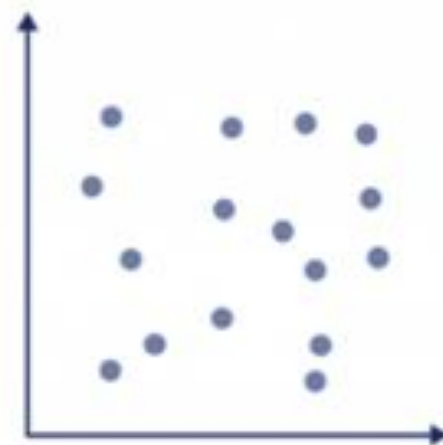
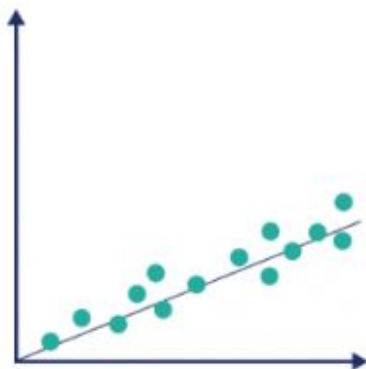
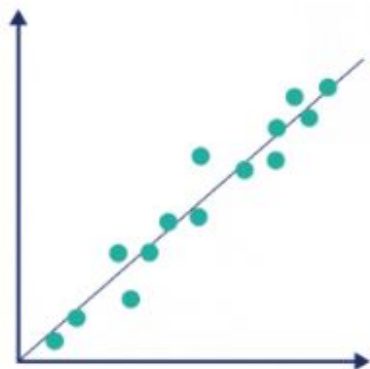
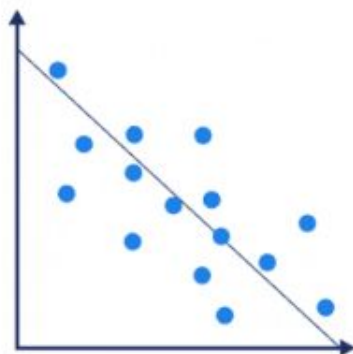
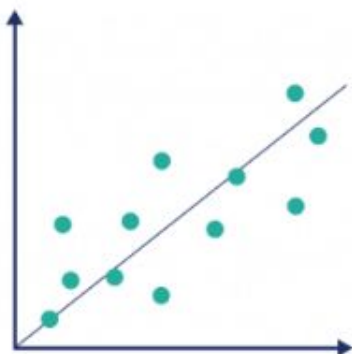
- It as "**Normalized Covariance.**" It's covariance divided by the product of standard deviations to remove units, making it comparable across different datasets.

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Examples

- **Scenario A ($r \approx 0.9$):** *Study Hours vs. Exam Scores.* As study time increases, scores almost always go up. A strong, positive trend.
- **Scenario B ($r \approx -0.8$):** *Car Age vs. Resale Value.* Generally, as a car gets older, its value drops significantly. A strong, negative trend.
- **Scenario C ($r \approx 0$):** *Shoe Size vs. Intelligence Quotient (IQ).* Knowing someone's shoe size tells you absolutely nothing about their IQ. No trend.
- **Workign example**
 - **Data:** $X=[1,2,3]$ and $Y=[2,4,6]$



- **The Correlation Spectrum**

- **Key Lesson:** As $|r|$ approaches 1, the points cluster more tightly around a straight line. As $|r|$ approaches 0, the points become a "cloud" or "blob" with no discernible direction.

- **The "Linear Only" Trap**

- Parabolic Curve $\Rightarrow r = 0$
- **Key Lesson:** Pearson only measures **linear** strength. A perfect relationship can exist, but if it isn't a straight line, Pearson will fail to detect it.

- **Slope is not Correlation**

CATEGORICAL CORRELATION, HYPOTHESIS TESTING AND CHI-SQUARED TEST

Introduction to Categorical Correlation

- **Concept:** Standard correlation (like Pearson's r) works only for numerical data. For **categorical** (nominal) data, we use the Chi-Square Test to see if two variables are "associated".
- **The Question:** Does knowing the value of one variable (e.g., *Gender*) help you predict the other (e.g., *Preferred Product Type*)?
- **Hypotheses:**
 - **H_0 (Null):** The variables are **Independent** (No correlation).
 - **H_a (Alternative):** The variables are **Dependent** (They are correlated).
 - Relation to A/B testing

Few more categorical examples

Categorical Variables

- Gender :
 - Male, Female
- Preferred NewsPaper :
 - TimesOfIndia, HindustanTimes, DH, ET, WSJ etc.
- Frequency of Television
 - Daily, few times a week, rarely, never
- Highest Education Level
 - School, College, Graduate, Masters, Ph.D.

Correlation Among Categorical Variables

- **Hypotheses:**
 - ***H₀* (Null):** The variables are **Independent** (No correlation).
 - ***H_a*(Alternative):** The variables are **Dependent** (They are correlated).
- **Possible Correlations**
 1. Gender [?] Preferred NewsPaper
 2. Gender [?] Frequency of Television
 3. Gender [?] Highest Education Level
 4. Highest Education Level [?] Preferred NewsPaper
 5. Highest Education Level [?] Frequency of Television
 6. Preferred NewsPaper [?] Frequency of Television

The Core Logic – Observed vs. Expected

- **Null Hypothesis**
 - Variable 1 and Variable 2 are not correlated
- **Observed (O):**
 - The actual frequency counts you collected.
- **Expected (E):**
 - The counts we would see if there were **zero** correlation.
- **Intuition:**
 - If the gap between "What we see" (O) and "What we expect" (E) is large, the variables are likely correlated.

The Mathematical Formulas

- **Chi-Square Statistic:**

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O = Observed frequency
- E = Expected frequency

- **Calculating Expected Value (E)**

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Interpreting Chi Square

- The **raw chi-square value alone** is misleading because:
 - It increases with sample size
 - It depends on table dimensions
 - Large chi-square $\neq$ strong association
- *Instead, we use **standardized effect size measures** that give us correlation-like strength indicators (0 to 1 scale).*

Cramer's V

$$V = \sqrt{\frac{\chi^2}{n \times (k - 1)}}$$

- $k = \min(\text{number of rows, number of columns})$
 - $n = \text{total sample size}$
- Interpretation
 - 0.10 = Weak
 - 0.30 = Moderate
 - 0.50 = Strong

Example

- Lets check for any correlation between variables ***Gender*** and ***Highest Education Level***
- Null Hypothesis : They arent correlated

	Female	Male	Row Total
School	6	7	13
College	13	16	29
Graduate	16	15	31
Masters	8	11	19
Column			
Total	43	49	92

Work Out Example

- Excel Sheets